

CCNA · ROUTING & SWITCHING LAB REPORT

STATIC ROUTING.

Manually engineering inter-network reachability across a three-router topology — from a routed subnet no one can reach, to full end-to-end connectivity with zero dynamic protocols involved.

VERIFIED PACKET PATH · WIN-1 → R1 → R2 → R3 → WIN-2



● R1 ↔ R2 link up ● R2 ↔ R3 link up ● 4/4 static routes active ● 0% end-to-end packet loss

TOPOLOGY

3 Routers · 2 Hosts

ROUTING METHOD

Static (ip route)

ADDRESSING

Classful /8

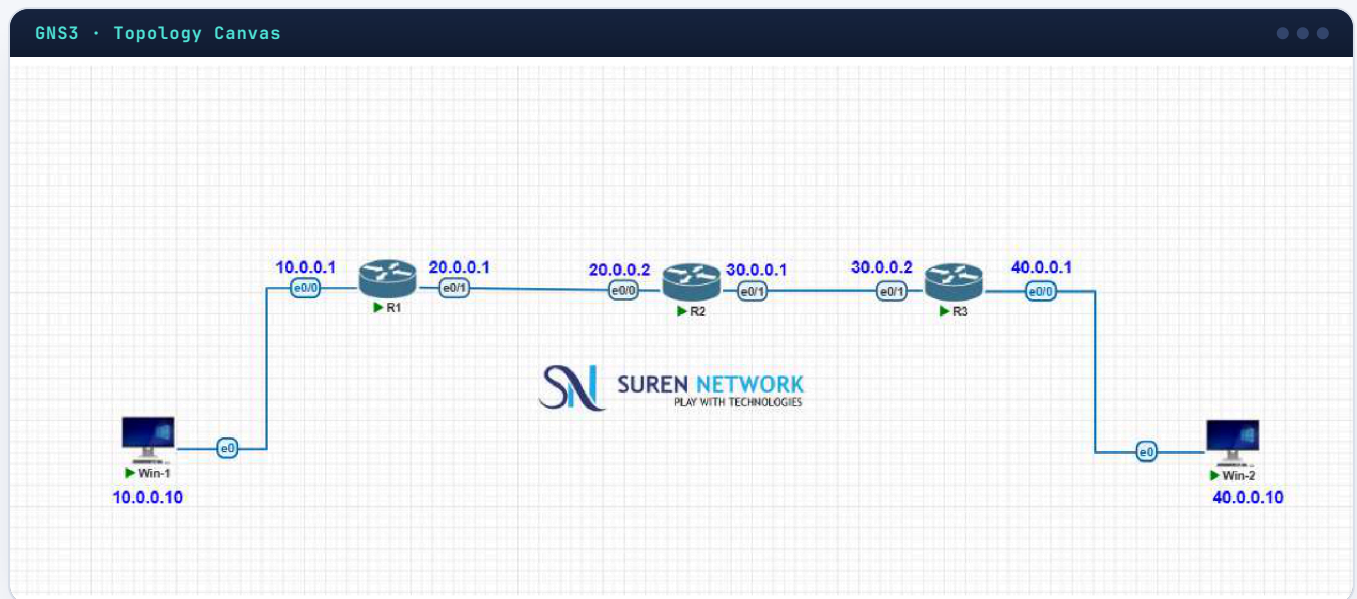
PLATFORM

Cisco IOS · GNS3

Objective & Topology

Two isolated LANs — **Win-1** on **10.0.0.0/8** and **Win-2** on **40.0.0.0/8** — sit three router-hops apart with no dynamic routing protocol enabled anywhere. The objective is to configure **manual static routes** on **R1**, **R2**, and **R3** so that both end hosts can reach each other, then verify the path hop-by-hop with ping and show ip route.

Network Diagram



What this lab proves

SKILL 01

Interface addressing & activation on Cisco IOS

SKILL 02

Diagnosing an unreachable destination

SKILL 03

Writing & verifying static routes end-to-end

Devices in scope

DEVICE	ROLE	INTERFACES USED	NOTES
R1	Edge router – Win-1 side	Ethernet0/0, Ethernet0/1	Gateway for 10.0.0.0/8
R2	Transit / core router	Ethernet0/0, Ethernet0/1	No directly attached hosts
R3	Edge router – Win-2 side	Ethernet0/0, Ethernet0/1	Gateway for 40.0.0.0/8
Win-1 / Win-2	End hosts	NIC (e0)	Windows XP test clients

Every link in the topology gets its own classful /8 network. Four networks in total — one per router-to-router or router-to-host segment — give every hop a distinct, easy-to-trace address space before a single route is written.

Full addressing table

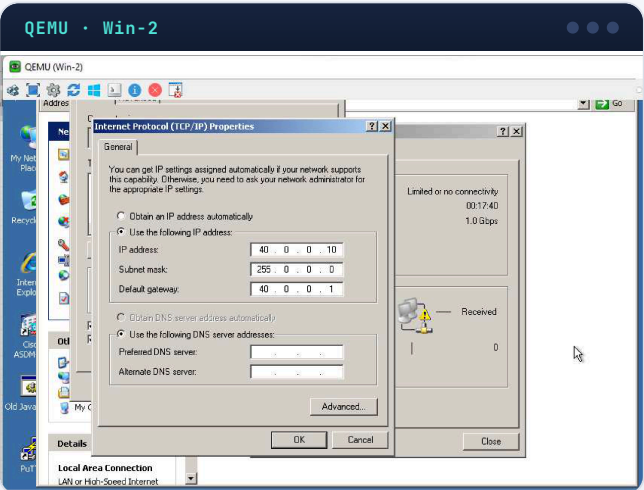
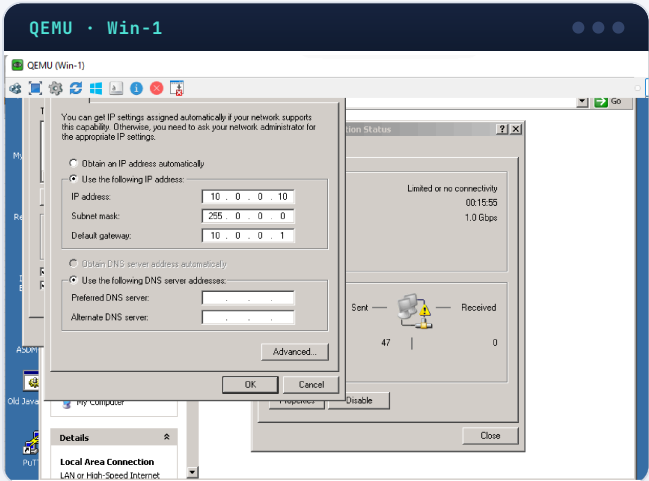
NETWORK	DEVICE	INTERFACE	IP ADDRESS	SUBNET MASK
NET 1	R1	Ethernet0/0	10.0.0.1	255.0.0.0
HOST	Win-1	NIC	10.0.0.10	255.0.0.0
NET 2	R1	Ethernet0/1	20.0.0.1	255.0.0.0
NET 2	R2	Ethernet0/0	20.0.0.2	255.0.0.0
NET 3	R2	Ethernet0/1	30.0.0.1	255.0.0.0
NET 3	R3	Ethernet0/1	30.0.0.2	255.0.0.0
NET 4	R3	Ethernet0/0	40.0.0.1	255.0.0.0
HOST	Win-2	NIC	40.0.0.10	255.0.0.0

Default gateways

WIN-1 GATEWAY
10.0.0.1 → R1

WIN-2 GATEWAY
40.0.0.1 → R3

Host TCP/IP configuration



Interface Configuration — R1

Each router interface is addressed manually and brought up with `no shutdown`. `show ip int brief` confirms line protocol is **up/up** before moving to the next device. R1 carries two interfaces — one facing Win-1, one facing the core toward R2.

R1 — Ethernet0/0 & Ethernet0/1

```
R1(config)# interface ethernet0/0
R1(config-if)# ip address 10.0.0.1 255.0.0.0
R1(config-if)# no shutdown
! second interface toward the core
R1(config)# interface ethernet0/1
R1(config-if)# ip address 20.0.0.1 255.0.0.0
R1(config-if)# no shutdown
```

Console verification

```
R1
R1#config term
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#interface ethernet0/0
R1(config-if)#ip add 10.0.0.1 255.0.0.0
R1(config-if)#no sh
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
R1(config)#exit
R1#
*Aug 23 06:09:49.710: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip int br
Interface IP-Address OK? Method Status Protocol
Ethernet0/0 10.0.0.1 YES manual up up
Ethernet0/1 unassigned YES TFTP down down
Ethernet0/2 unassigned YES TFTP down down
Ethernet0/3 unassigned YES TFTP down down
R1#
```

```
R1
R1#
R1#
*Aug 23 06:10:41.400: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#
R1#config term
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface ethernet0/1
R1(config-if)#ip add 20.0.0.1 255.0.0.0
R1(config-if)#exit
*Aug 23 06:11:31.673: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#exit
R1#
*Aug 23 06:11:47.780: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip int br
Interface IP-Address OK? Method Status Protocol
Ethernet0/0 10.0.0.1 YES manual up up
Ethernet0/1 20.0.0.1 YES manual up up
Ethernet0/2 unassigned YES TFTP down down
Ethernet0/3 unassigned YES TFTP down down
R1#
```

Both interfaces show **up/up** in `show ip int brief` — R1 is ready to be pointed at a next hop once static routes are introduced.

Interface Configuration — R2 & R3

R2 sits in transit with no directly attached hosts; R3 terminates the topology on the Win-2 side. Both are addressed and verified the same way as R1.

R2 — Ethernet0/0 & Ethernet0/1

```
R2(config)# int ethernet0/1
R2(config-if)# ip address 30.0.0.1 255.0.0.0
R2(config-if)# no shutdown
! Ethernet0/0 (20.0.0.2) was already active toward R1
```

R2 · Console — show ip int brief

```
R2#
*Aug 23 06:14:07.921: XSYS-5-CONFIG_I: Configured from console by console
R2#
R2#config term
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int ethernet0/1
R2(config-if)#ip add 30.0.0.1 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#exit
R2#
*Aug 23 06:15:41.421: XSYS-5-CONFIG_I: Configured from console by console
R2#sh ip int br
Interface      IP-Address      OK? Method Status  Prot
Ethernet0/0    20.0.0.2        YES manual up      up
Ethernet0/1    30.0.0.1        YES manual up      up
Ethernet0/2    unassigned      YES TFTP  down    down
Ethernet0/3    unassigned      YES TFTP  down    down
R2#
```

R3 — Ethernet0/1 & Ethernet0/0

```
R3(config)# int ethernet0/1
R3(config-if)# ip address 30.0.0.2 255.0.0.0
R3(config-if)# no shutdown
! LAN-facing interface toward Win-2
R3(config)# int ethernet0/0
R3(config-if)# ip address 40.0.0.1 255.0.0.0
R3(config-if)# no shutdown
```

R3 · Console — show ip int brief

```
R3#config term
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int ethernet0/1
R3(config-if)#ip add 30.0.0.2 255.0.0.0
R3(config-if)#no sh
R3(config-if)#exit
R3(config)#exit
R3#
*Aug 23 06:17:45.762: XSYS-5-CONFIG_I: Configured from console by console
R3#
R3#config term
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int ethernet0/0
R3(config-if)#ip add 40.0.0.1 255.0.0.0
R3(config-if)#no sh
R3(config-if)#
*Aug 23 06:18:32.902: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R3(config-if)#exit
R3(config)#exit
R3#
*Aug 23 06:18:39.581: XSYS-5-CONFIG_I: Configured from console by console
R3#sh ip int br
Interface      IP-Address      OK? Method Status  Protocol
Ethernet0/0    40.0.0.1        YES manual up      up
Ethernet0/1    30.0.0.2        YES manual up      up
Ethernet0/2    unassigned      YES TFTP  down    down
Ethernet0/3    unassigned      YES TFTP  down    down
R3#
```

Interface Status — Summary

Before a single route is written, it's worth confirming every routed interface across all three routers is physically and logically healthy.

Interface status — all three routers

ROUTER	INTERFACE	IP ADDRESS	STATUS	PROTOCOL
R1	Ethernet0/0	10.0.0.1	UP	UP
R1	Ethernet0/1	20.0.0.1	UP	UP
R2	Ethernet0/0	20.0.0.2	UP	UP
R2	Ethernet0/1	30.0.0.1	UP	UP
R3	Ethernet0/1	30.0.0.2	UP	UP
R3	Ethernet0/0	40.0.0.1	UP	UP

All six routed interfaces report **up/up** — Layer 1 and Layer 2 are healthy on every link before any Layer 3 routing logic is introduced. That isolates the next diagnosis to routing alone.

Why this checkpoint matters

A down interface and a missing route produce the exact same symptom to an end user — "it doesn't work." Confirming Layer 1/2 health first means any failure seen next can be attributed cleanly to the routing table, not the wire.

The Problem — No Routes Yet

With every interface up, Win-1 still cannot reach Win-2. Each router only knows about its **own directly connected** networks — R1 has never heard of 40.0.0.0/8, so it has nowhere to forward the packet.

Host ipconfig

```
Win-1 · cmd.exe

C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 10.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1

C:\Documents and Settings\user>
```

R1's view of the network

```
R1 · show ip route

R1#
R1#sh ip int br
Interface      IP-Address      OK? Method Status      Protocol
Ethernet0/0    10.0.0.1        YES manual up          up
Ethernet0/1    20.0.0.1        YES manual up          up
Ethernet0/2    unassigned      YES TFTP  down        down
Ethernet0/3    unassigned      YES TFTP  down        down
R1#
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - OOR, P - periodic downloaded static route, H - NHRP, I - IGRP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
C       20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
R1#
```

Result: ping fails

```
Win-1 · ping 40.0.0.10

C:\WINDOWS\system32\cmd.exe
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 10.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1

C:\Documents and Settings\user>ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:

Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.
Reply from 10.0.0.1: Destination host unreachable.

Ping statistics for 40.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>
```

Destination host unreachable — R1's routing table contains only 10.0.0.0/8 and 20.0.0.0/8. Without a route to 40.0.0.0/8, R1 has no next hop and drops the packet immediately. This is the exact gap static routing is about to close.

Static Route Configuration

Each router only needs a route toward networks it **isn't** directly connected to. R2 needs two (one in each direction); the edge routers R1 and R3 need just one apiece, pointed at R2.

```
! R1 - route to Win-2's network via R2
R1(config)# ip route 40.0.0.0 255.0.0.0 20.0.0.2

! R2 - routes back to Win-1 and forward to Win-2
R2(config)# ip route 10.0.0.0 255.0.0.0 20.0.0.1
R2(config)# ip route 40.0.0.0 255.0.0.0 30.0.0.2

! R3 - route to Win-1's network via R2
R3(config)# ip route 10.0.0.0 255.0.0.0 30.0.0.1
```

Routing table on R1 after the fix

```
R1 - show ip route
R1#
C 20.0.0.0/8 is directly connected, Ethernet0/1
L 20.0.0.1/32 is directly connected, Ethernet0/1
R1#config term
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip route 40.0.0.0 255.0.0.0 20.0.0.2
R1(config)#exit
R1#
*Aug 23 06:32:32.112: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/8 is directly connected, Ethernet0/0
L 10.0.0.1/32 is directly connected, Ethernet0/0
C 20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 20.0.0.0/8 is directly connected, Ethernet0/1
L 20.0.0.1/32 is directly connected, Ethernet0/1
S 40.0.0.0/8 [1/0] via 20.0.0.2
R1#
```

Static routing table — full summary

ROUTER	DESTINATION	MASK	NEXT HOP
R1	40.0.0.0	255.0.0.0	20.0.0.2 (R2)
R2	10.0.0.0	255.0.0.0	20.0.0.1 (R1)
R2	40.0.0.0	255.0.0.0	30.0.0.2 (R3)
R3	10.0.0.0	255.0.0.0	30.0.0.1 (R2)

R1 now carries a default-free but complete table: its own two connected networks, plus a single static route covering everything beyond R2.

Routing Tables — R2 & R3

R2 is the only router carrying two static entries — one back toward Win-1, one forward toward Win-2 — since it has no directly connected hosts of its own.

Routing table on R2

```
R2 • show ip route

R2#
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.1/32 is directly connected, Ethernet0/1
R2#
R2#config term
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
R2(config)#ip route 40.0.0.0 255.0.0.0 30.0.0.2
R2(config)#exit
R2#
*Aug 23 06:39:23.849: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
C    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    20.0.0.0/8 is directly connected, Ethernet0/0
L    20.0.0.2/32 is directly connected, Ethernet0/0
C    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.1/32 is directly connected, Ethernet0/1
S    40.0.0.0/8 [1/0] via 30.0.0.2
R2#
```

Routing table on R3

```
R3 • show ip route

R3#
R3#config term
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
R3(config)#exit
R3#
*Aug 23 06:49:36.909: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PFR

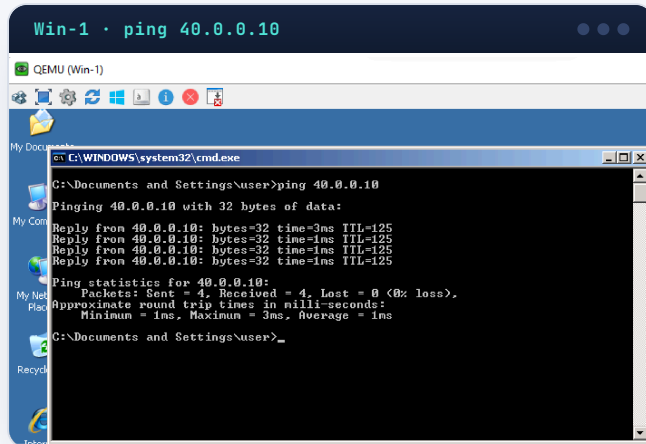
Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 30.0.0.1
C    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.2/32 is directly connected, Ethernet0/1
C    40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    40.0.0.0/8 is directly connected, Ethernet0/0
L    40.0.0.1/32 is directly connected, Ethernet0/0
R3#
```

Every router now has a path — direct or static — to both 10.0.0.0/8 and 40.0.0.0/8. The topology is ready for an end-to-end test.

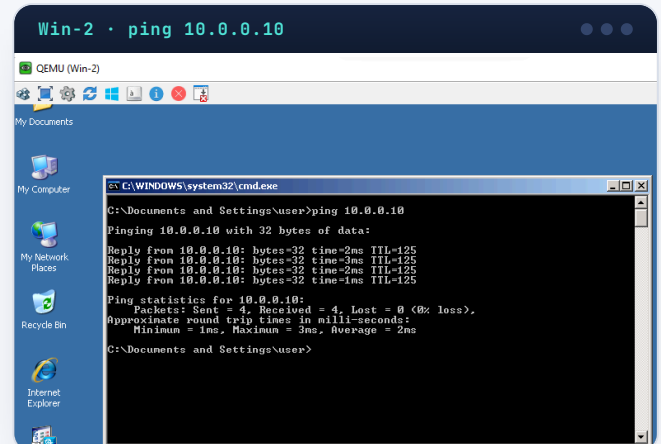
End-to-End Connectivity

With routes in place on all three routers, both hosts ping across the full topology with **0% packet loss** — confirming the return path works just as well as the forward path.



0% LOSS

TTL 125

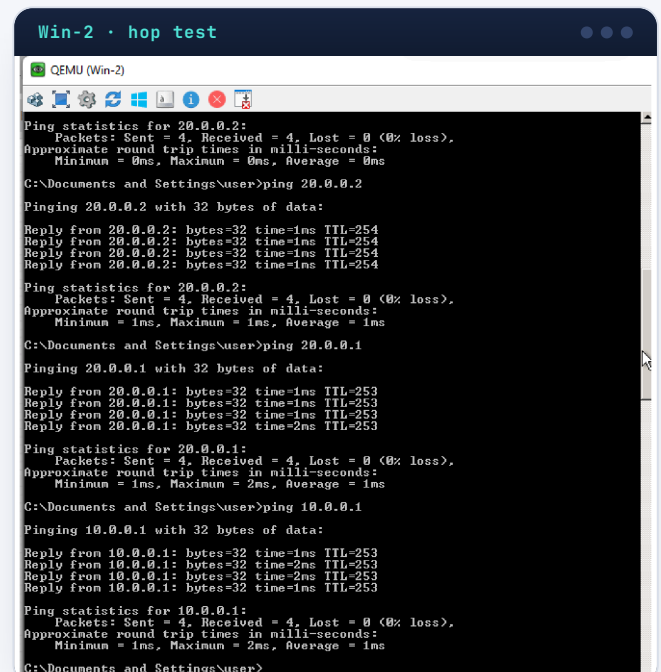
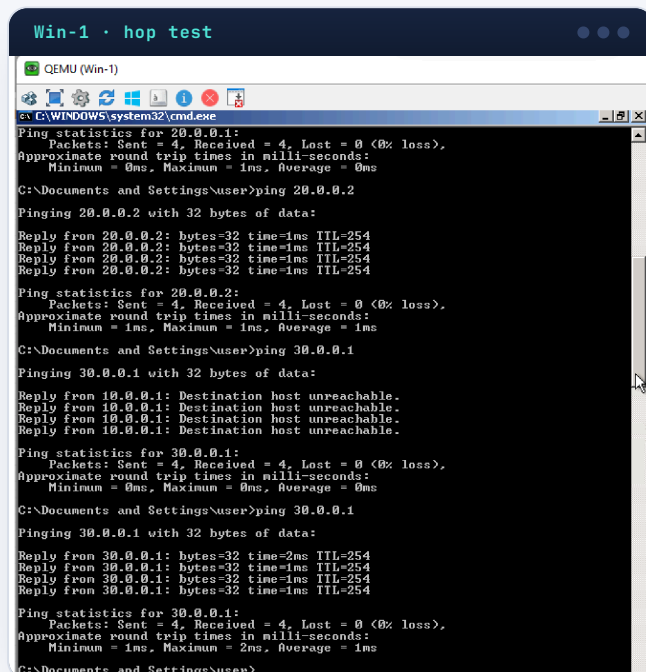


0% LOSS

TTL 125

Hop-by-hop trace — every interface along the path

Pinging each router interface in turn from both hosts confirms the packet really is traversing R1 → R2 → R3, not taking a shortcut.



Conclusion & Key Takeaways

Static routing gave full, verifiable control over how traffic moves between 10.0.0.0/8 and 40.0.0.0/8 — three routers, four routes, zero dynamic protocols. The exercise reinforced how a router's forwarding decision is only ever as good as the routes sitting in its table.

TAKEAWAY 01

A router with no matching route drops the packet — “up/up” interfaces alone don’t guarantee reachability.

TAKEAWAY 02

Transit routers need routes in both directions — R2 required two static entries, the edges only one.

TAKEAWAY 03

show ip route and hop-by-hop pings are the fastest way to prove a path, not just assume it.

Command reference — full config recap

```
R1: ip route 40.0.0.0 255.0.0.0 20.0.0.2
R2: ip route 10.0.0.0 255.0.0.0 20.0.0.1 ip route 40.0.0.0 255.0.0.0 30.0.0.2
R3: ip route 10.0.0.0 255.0.0.0 30.0.0.1
```